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The impact of non-stiff fuel injection on the acoustic response of a partially premixed hydrogen flame

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ABSTRACT

Micro-mix combustion has been proposed to mitigate flashback and maintain low NOx in fuel-flexible gas turbines. In such partially premixed configurations, both velocity and pressure fluctuations can induce equivalence ratio fluctuations near the fuel injector. Once the pressure oscillation reaches a significant level, the assumption of stiff fuel injection will be compromised, resulting in fluctuations in fuel mass flow. This study investigates the influence of non-stiff injection, where the fuel mass flow can be modulated by pressure fluctuations, on the dynamic response of a partially premixed hydrogen flame. A model incorporating the acoustic impedance of a circular aperture in the fuel line is developed to predict the evolution of equivalence ratio fluctuations and validated using time-resolved ultraviolet absorption spectroscopy. The model reproduces the measured gain under non-stiff injection, whereas assuming stiff injection underestimates it. Flame transfer functions (FTF) based on OH* chemiluminescence exhibit gain modulation over frequency, arising from constructive and destructive interferences between velocity- and equivalence ratio-driven oscillations, with Strouhal number scaling effectively collapsing the phase and the positions of peaks and troughs in the FTF gain. However, variations in the FTF gain across different mean flow velocities suggest an additional contribution of pressure-induced equivalence ratio fluctuations. To identify this effect, an alternative FTF is introduced using the fuel injection velocity as the reference input. The resultant collapse confirms the role of pressure-induced equivalence ratio fluctuations. Finally, phase-averaged OH* analysis reveals distinct responses along the flame. These findings offer valuable insights for the design of future hydrogen-fueled combustion systems.

1. Introduction

In the power generation industry, the transition from natural gas to renewable fuels has heightened the demand for fuel-flexible gas turbines capable of handling peak loads and the variability of wind and solar energy [1,2]. Hydrogen, recognized as a promising alternative decarbonized fuel, has received considerable attention [3–5]. However, the shift from natural gas to hydrogen in modern well-established gas turbines, which rely on lean premixed combustion to reduce NOx emissions, increases the risk of flashback, potentially causing catastrophic damage to the structure [6–10]. This necessitates alternative strategies to enable safe and stable hydrogen combustion.

The micro-mix concept has recently gained renewed attention for the burning of pure hydrogen [11–18]. Instead of achieving a fully premixed fuel–air mixture far upstream of the flame, this strategy involves injecting fuel near the exit of the burner, allowing for rapid mixing with the incoming air flow. Furthermore, it utilizes a series of miniaturized flames, reducing residence time in high-temperature regions and consequently reducing NOx emissions [11]. Among various configurations, fuel injection through some small jet-in-crossflow arrangement has received particular interest [12,19–21], as it effectively mitigates flashback by preventing reactive mixtures from forming upstream of the injection point. However, this leads to insufficient mixing and is thus often referred to as a partially premixed (or technically premixed) system since the non-uniformity of the equivalence ratio is still present at the flame.

In partially premixed flames, both velocity and equivalence ratio fluctuations contribute to unsteady heat release, which in turn drives thermoacoustic instability. This instability is a flow instability that arises due to two-way coupling between acoustic waves and unsteady heat release rate [22], causing large amplitude pressure fluctuations, and in extreme cases, catastrophic hardware failure [23]. Previous studies have shown that acoustic perturbations at the fuel injection

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location can induce temporal fluctuations in the equivalence ratio through a mechanism commonly referred to as a mode conversion process [24–26]. For simplicity, the fuel injector is commonly assumed to be choked and acoustically stiff in theoretical modeling [25], where the fuel mass flow is assumed to be constant. Experimental investigations must therefore be carefully designed and precisely controlled to satisfy these assumptions.

However, in some experimental rigs and real applications, the fuel injector may not be strictly choked [27]. In these cases, if the mean pressure drop across the fuel injector is not much larger than the pressure oscillation amplitude inside the combustor, as could be the technically practical configuration, non-stiff fuel injection needs to be considered. For example, observations by Bluemner et al. [26] reveal that when a pressure antinode forms at the fuel injector, large pressure fluctuations can compromise the injector's acoustic stiffness. Earlier foundational work by Lieuwen and Zinn [24] investigated the impact of pressure oscillations on fuel injector mass flow rate fluctuations, showing that injector dynamics can significantly affect combustor stability if the injector is not choked. Further, Huber and Polifke [28] examined the role of fuel supply impedance in lean premixed combustors, demonstrating that non-stiff fuel injection can have a substantial impact on heat release fluctuations and, consequently, the thermoacoustic stability of the system.

While the aforementioned mechanisms highlight the complexity introduced by equivalence ratio fluctuations, the flame dynamics can still be effectively characterized by a single-input–single-output (SISO) flame transfer function (FTF) approach if the fuel line is stiff and the burner is acoustically compact. These FTFs characterize the relation between upstream velocity fluctuations and the resulting overall heat release rate fluctuations, which already incorporate the effect of equivalence ratio fluctuations. This approach offers a practical framework for analyzing and predicting thermoacoustic response, particularly when the burner is acoustically compact relative to the wavelengths of interest. For example, Kim et al. [29] experimentally investigated a swirl-stabilized, partially premixed hydrogen-enriched flame and showed that flame response is strongly influenced by the phase difference between velocity and equivalence ratio fluctuations, with in-phase forcing leading to nonlinear behavior such as local extinction and elevated FTF gain. They also found that fuel injector impedance slightly affected the phase difference between choked and unchoked injection conditions. Moon et al. [21] experimentally studied FTFs in fully and partially premixed hydrogen flames while ensuring acoustically stiff fuel injection via carefully controlled nitrogen dilution to maintain high injection speed. Their results revealed a clear gain modulation phenomenon in partially premixed configurations, caused by constructive and destructive interference between velocity- and equivalence ratio-induced heat release rate fluctuations. Similar gain modulation behavior has been observed in premixed swirling flames [30–32], where it was attributed to vortex shedding or swirling number fluctuation from bluff bodies or swirlers. These aforementioned findings highlight the critical role of the interference between velocity and equivalence ratio fluctuations in partially premixed systems. However, comprehensive physical explanation or quantitative prediction of either the frequencies or values of the peaks and troughs of the FTF gain has not been provided. To address this, some studies have adopted multi-input–single-output (MISO) frameworks that separate the contributions of velocity and equivalence ratio fluctuations. These have been explored through both numerical simulations [33–35] and experiments [36,37], offering a more detailed understanding of how different perturbation mechanisms interact to shape flame dynamics.

Although FTF-based studies have been extensively conducted for hydrocarbon fuels, investigations focusing on pure hydrogen–air flames — particularly under partially premixed conditions with non-stiff (i.e., compliant) fuel injection — remain limited. The present study aims to experimentally characterize the flame response of a partially premixed hydrogen burner with acoustically non-stiff fuel injection, a regime in which the conventional assumption of constant fuel mass flow is no longer valid, particularly once thermoacoustic instability develops and pressure oscillations reach significant amplitudes. The paper is organized as follows: First, the experimental setup and the optical diagnostics methodologies are described in the next section. In Section 3, a model is developed to quantify the mode conversion (equivalence ratio fluctuation with a given inlet velocity fluctuation) in the vicinity of the fuel injection and is validated against the equivalence ratio measurements at the burner outlet. Section 4 presents the FTFs based on OH* measurements across a range of air flow rates and global equivalence ratios. This section also explores Strouhal number scaling and introduces an alternative FTF formulation to elucidate the dominant physical mechanisms. Finally, a spatiotemporal analysis of OH* chemiluminescence is conducted to further reveal the underlying physical mechanisms responsible for the flame dynamics under simultaneous velocity and equivalence ratio perturbations. This research provides new insights into the dynamic response of pure hydrogen flames under partially premixed, acoustically non-stiff fuel injection conditions, and offers valuable insights for the design of future hydrogen-fueled combustion systems.

2. Experimental methods

2.1. Experimental setup

As shown in Fig. 1(a), the present experimental rig consists of an air inlet, a loudspeaker unit, an acoustic test pipe, and a micro-mix burner. Both the air inlet and the acoustic test pipe were equipped with a flow straightener to eliminate large-scale vortical structures and ensure uniform flow (confirmed by hot wire measurement of the velocity profile conducted at the outlet of the acoustic test pipe). Acoustic perturbations were generated using two loudspeakers mounted upstream of the test pipe. To enable accurate reconstruction of the acoustic velocity field, five microphone ports were distributed non-uniformly along the test pipe, as shown in Fig. 1(b). The multi-microphone method (MMM) [38] was used to reconstruct the 1-D acoustic field in the test pipe, which was equipped with 4 flush-mounted 1/4″ microphones (GRAS 64BD-FV). The velocity at the reference plane ($x = 0$) was taken as the reference velocity u_r. It has been checked that the present MMM can achieve high accuracy of both the amplitude (within 1% relative error) and phase (within 1°) of the plane acoustic wave reflection coefficient inside the test pipe for the considered frequency range [39].

The burner employed in this study is a laboratory-scaled micro-mix burner with an inner diameter (D) of 6.3 mm, as shown in Fig. 1(c). Hydrogen was injected radially into the airflow through four equally spaced holes. Each hole has an inner diameter of 0.4 ± 0.01 mm, and is positioned at $L_{\mathrm{inj}} = 96.7$ mm upstream of the burner exit. A centerbody-less swirler with a swirl number of 0.43 was designed to enhance the mixing process, while a downstream flow straightener, as shown in Fig. 1(e), suppressed vortical and swirling flow structures before the flow exits the burner.

2.2. The measurement of equivalence ratio fluctuations

The relation between the upstream velocity perturbations and the equivalence ratio fluctuations at the burner outlet is characterized using a transfer function T_ϕ, defined as

$$T_\phi(\omega) = \frac{\hat{\phi}(\omega)/\bar{\phi}}{\hat{u}_r(\omega)/\bar{u}_r}, \tag{1}$$

where ϕ denotes the equivalence ratio at the burner outlet. The notation $\widehat{[\ ]}$ refers to the Fourier amplitude, while $\overline{[\ ]}$ denotes the time-averaged value.

To determine the time-resolved ϕ, broadband ultraviolet (UV) absorption spectroscopy was employed, as shown in Fig. 1(f). Sulfur dioxide (SO_2) was selected as a tracer gas at a 10% mole fraction

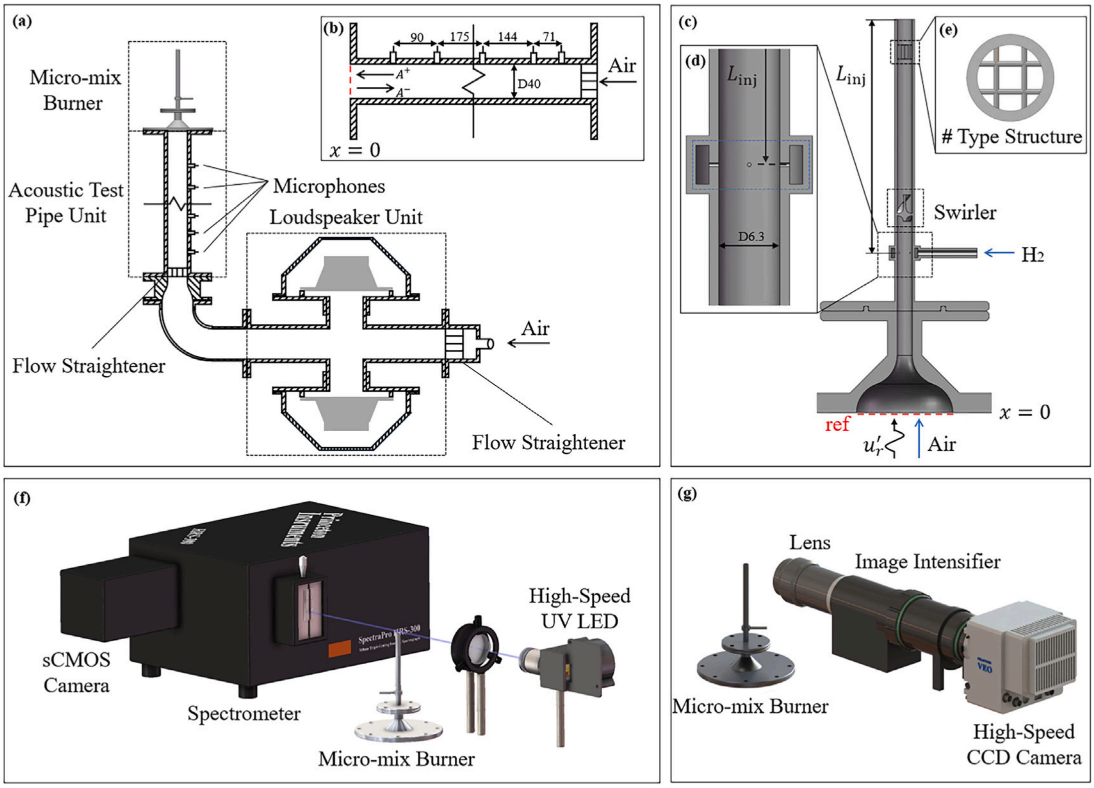

Fig. 1. Experimental setup. (a) Overview of the experimental rig. (b) Detailed view of the acoustic test pipe. (c) Overview of the micro-mix burner. (d) Detailed view of the fuel injection. (e) Top view of a flow straightener nearby the burner exit. (f) Time-resolved equivalence ratio measurement system. (g) Time-resolved OH* intensity measurement system.

in the fuel stream due to its strong absorption characteristics near 305 nm [40]. A high-speed UV LED centered at 305 nm served as the light source. It was operated at 1 kHz with a 100 μs pulse width and synchronized with an sCMOS camera running in rolling gate mode. The collimated UV beam was focused onto a 2 mm spot located 2 mm above the burner outlet and directed into a spectrometer (1200 g/mm). Variations in the SO_2 absorption spectrum were used to determine its line-of-sight integrated concentration. The corresponding hydrogen concentration was inferred based on this measurement. Further details of this diagnostic system, which was also used in our prior study of parasite-velocity decoupled equivalence ratio modulations, can be found in [41] and supplementary materials.

2.3. Flame transfer function measurement

The relation between the upstream velocity perturbations and the overall flame response is described by the FTF expressed by

$$F(\omega) = \frac{\hat{I}(\omega)/\bar{I}}{\hat{u}_r(\omega)/\bar{u}_r}, \tag{2}$$

where I represents the OH* chemiluminescence intensity, and was used as an indicator of the overall heat release rate fluctuations in this study. The emission signal was filtered through a 310 ± 10 nm bandpass filter and recorded using a high-speed camera (Phantom VEO 1310L) equipped with a UV lens and a high-speed image intensifier (Lambert HiCATT), as shown in Fig. 1(g). The imaging system operated at a repetition rate of 10 kHz, providing sufficient temporal resolution to capture the relevant flame dynamics.

It should be noted that although the OH* chemiluminescence intensity is a widely used indicator of heat release rate in fully premixed flames [42], its reliability diminishes under partially premixed conditions with equivalence ratio fluctuations [43,44], due to the direct dependence of OH* emission intensity on the local equivalence ratio [45,46]. This may lead to an overprediction of the influence of equivalence ratio fluctuations. A detailed quantitative assessment of chemiluminescence-based heat release measurements is beyond the scope of the present study and is left for future work. Nevertheless, owing to its optical accessibility for the full field and diagnostic simplicity in practical combustion systems, it remains a useful indication of unsteady heat release. It is worth mentioning that the method based on acoustic transfer matrix could be a good alternative way — it provides information about the heat release rate (as the sound source) and thus the FTF [21,38,47], but could not provide optical information about the flame dynamics. For these reasons, OH* intensity is employed in the present work to represent the heat release rate fluctuations.

2.4. Operating conditions

Two different fuel injection schemes were employed in this study, as illustrated in Fig. 2, each tailored to a specific measurement objective. The fuel trace scheme was used to measure the equivalence ratio fluctuations, and the partially premixed scheme was employed for FTF measurements. In contrast to systems with stiff fuel injection, the present setup did not particularly control the fuel injection velocity, and it is operated at a relatively low nominal Mach number (∼0.15).

Operating conditions of partially premixed scheme are summarized in Table 1, covering different combinations of air flow rate ($\overline{V}_{\mathrm{Air}}$) and global equivalence ratio ($\bar{\phi}$) at ambient temperature and atmospheric pressure. Here, J denotes the momentum flux ratio between the fuel and air flow, Re is the Reynolds number based on the bulk velocity ($\bar{u}$) and inner diameter D, and $u'_{\mathrm{rms}}/\bar{u}$ represents the turbulence intensity. The fuel-trace scheme is based on the same partially premixed scheme as the baseline experiments, with the only modification being the addition of a small amount of SO_2 to the fuel flow for diagnostic purposes, while maintaining the same fuel injection velocity.

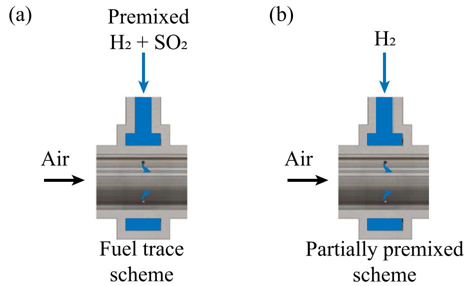

Fig. 2. Two different injection schemes in this study: (a) Fuel trace scheme in equivalence ratio measurement; (b) Partially premixed scheme in FTF measurement.

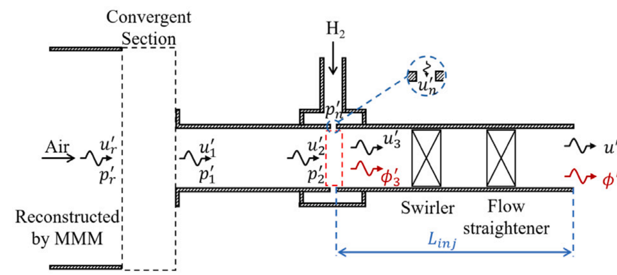

Fig. 3. Sketch of the generation of equivalence ratio fluctuation (ϕ') under acoustic forcing in the partially premixed burner of the present work.

Table 1
Operating points.

Parameters [Units]	Value
$\overline{V}_{\mathrm{Air}}$ [L/min]	17.96, 21.70, 25.44
$\bar{\phi}$ [−]	0.56, 0.66, 0.76
$\bar{u}$ [m/s]	11–19
J [−]	15–27
$u'_{\mathrm{rms}}/\bar{u}$ [−]	0.16–0.19
Re [−]	4000–6000
$\lvert\hat{u}_r/\bar{u}_r\rvert$ [−]	5% ± 0.5%

Equivalence ratio and OH* measurements were performed independently using separate diagnostic systems described earlier, but both were synchronized with microphone signals. To maintain linear flame response, the amplitude of the imposed velocity perturbations was controlled at $5\% \pm 0.5\%$ of the mean flow velocity. Due to limitations in the repetition rate of the equivalence ratio measurement system, the forcing frequency in the fuel trace scheme was varied from 60 to 360 Hz in 20 Hz increments. For FTF measurements, the forcing frequency was varied from 60 to 600 Hz in 20 Hz increments.

3. Transfer function of equivalence ratio fluctuations

3.1. Generation of equivalence ratio fluctuations

Fig. 3 shows a schematic of the burner configuration, highlighting the key perturbations relevant to the analysis that follows. Fluctuations of quantities about the time average are denoted with [ ]′. The generation of equivalence ratio fluctuations arises from the oscillations in the flow rate of both fuel ($\dot{m}_f$) and air ($\dot{m}_a$). This relation can be derived by linearizing the definition of equivalence ratio [24]:

$$\frac{\hat{\phi}_3}{\bar{\phi}_3} \approx \frac{\hat{\dot{m}}_f}{\bar{\dot{m}}_f} - \frac{\hat{\dot{m}}_a}{\bar{\dot{m}}_a} \approx \frac{\hat{u}_n}{\bar{u}_n} - \frac{\hat{u}_2}{\bar{u}_2}, \tag{3}$$

where ϕ_3 is the equivalence ratio just downstream of the fuel injection point, u_n denotes the fuel injection velocity, and u_2 is the airflow velocity immediately upstream of the fuel injection.

For an acoustically compact burner, where the acoustic wavelength is much larger than the physical length of the burner, the following approximation can be established:

$$\frac{\hat{u}_r}{\bar{u}_r} \approx \frac{\hat{u}_2}{\bar{u}_2}, \quad \hat{p}_r \approx \hat{p}_2. \tag{4}$$

To relate the pressure fluctuation $\hat{p}_2$ and the resulting injection velocity fluctuation $\hat{u}_n$, the classical acoustic impedance model for a circular aperture is employed, which takes the following form [48,49]:

$$\hat{p}_n - \hat{p}_2 = i\omega\bar{\rho}_n l_{\mathrm{eff}}\hat{u}_n + \bar{\rho}_n C_d \bar{u}_n \hat{u}_n, \tag{5}$$

where $\hat{p}_n$ is the pressure fluctuation in the fuel plenum, $\bar{\rho}_n$ is the mean fuel density, l_{eff} is the effective length of the orifice accounting for inertial effects, and C_d is the discharge coefficient. Given the small orifice size and the relatively high fuel injection velocity, the imaginary part of the RHS of Eq. (5) is significantly smaller than the real part of the RHS of Eq. (5), in the considered frequency range of the present study. As a result, the inertial contribution can be neglected, and the model simplifies to a quasi-steady relation between pressure and velocity fluctuations at the injector. The pressure fluctuation $\hat{p}_n$ in the fuel plenum is small in the present study, so it is assumed to be zero — this is validated by microphone measurements detailed in the supplementary materials. In this study, the discharge coefficient is fixed at $C_d = 0.8$, which is estimated with reference to the values reported in [50]. With these simplifications, the relation between the incoming perturbations and the resulting equivalence ratio fluctuations becomes:

$$\frac{\hat{\phi}_3}{\bar{\phi}_3} = -\underbrace{\frac{1}{J M_2 C_d}\frac{\hat{p}_2}{\bar{\rho}_2\bar{c}_2\bar{u}_2}}_{\text{pressure-driven}} - \underbrace{\frac{\hat{u}_2}{\bar{u}_2}}_{\text{velocity-driven}}, \tag{6}$$

where $J = \bar{\rho}_n\bar{u}_n^2/\bar{\rho}_2\bar{u}_2^2$ denotes the momentum flux ratio of jet-in-crossflow, and $M_2 = \bar{u}_2/\bar{c}_2$ is the Mach number of airflow immediately upstream of the injection site.

In the present study, the velocity fluctuations at the reference plane (u_r) are actively controlled by the input from the upstream loudspeaker, thereby imposing a known acoustic excitation, while the corresponding pressure fluctuations are inherently decided by the velocity fluctuations through the acoustic impedance of the burner. Fig. 4 compares the normalized pressure perturbation magnitude before and after scaling by the factor $1/(J M_2 C_d)$ with the normalized velocity perturbations at a global equivalence ratio $\bar{\phi} = 0.66$. The velocity perturbation remains nearly constant across the frequency range, while the normalized pressure exhibits a pronounced frequency dependence, with a clear peak near 200 Hz. The pressure-driven term is significantly amplified by this factor and becomes larger than the velocity-driven term around 200 Hz. As the bulk velocity increases, the factor $1/(J M_2 C_d)$ decreases, since the momentum flux ratio J is maintained constant at a fixed $\bar{\phi}$. Consequently, the amplification effect is more pronounced at lower bulk velocities. This behavior highlights the sensitivity of the pressure-driven response to the flow rate and plays a key role in determining the extent of pressure-driven fuel modulation in the following analysis.

To quantify the relation between the generated equivalence ratio fluctuations and the imposed velocity fluctuations at the reference plane, a model for the generation of equivalence ratio fluctuations, G_ϕ is introduced. This model is obtained by combining Eqs. (4)–(6) and is defined as:

$$G_\phi = \frac{\hat{\phi}_3/\bar{\phi}_3}{\hat{u}_r/\bar{u}_r} = g\left(\frac{-Z_r\bar{u}_r}{\bar{\rho}_n C_d \bar{u}_n^2} - 1\right), \tag{7}$$

where g is a modified scaling factor introduced to constrain the over-prediction behavior, possibly because of the damping effects near the jet-in-crossflow region. Similar overprediction behavior has been reported in [26]. A similar approach, involving the introduction of a

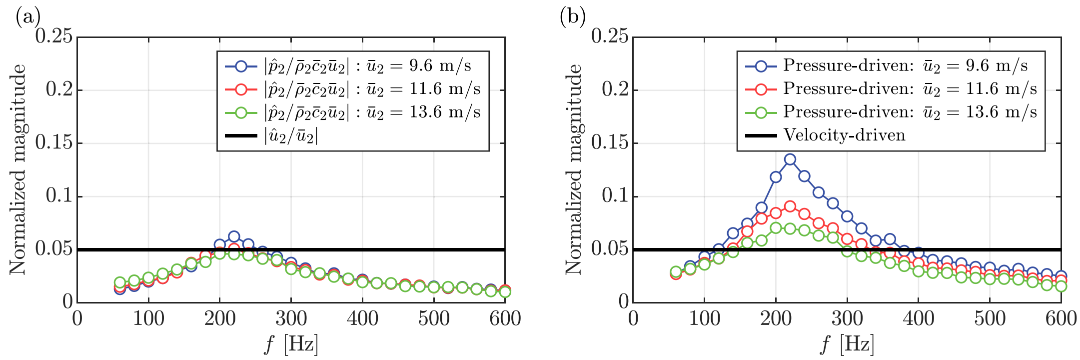

Fig. 4. Decomposition of the equivalence ratio fluctuations into pressure-driven and velocity-driven contributions for different bulk velocities at a fixed global equivalence ratio $\bar{\phi} = 0.66$. (a) Normalized magnitude of the unscaled pressure perturbation compared with the velocity perturbation. (b) Normalized magnitude of the pressure-driven contribution after scaling by the factor $1/(JM_2C_d)$, compared with the velocity-driven term. As $\hat{u}_2/\bar{u}_2$ is kept constant in the experiments, only one curve is shown.

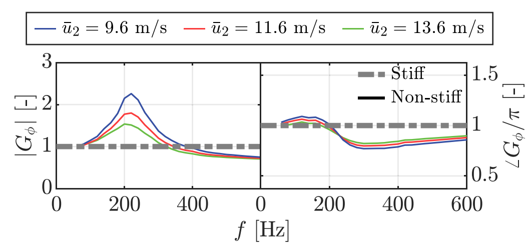

Fig. 5. Equivalence ratio fluctuation generation models G_ϕ for the cases with $\bar{\phi} = 0.66$. Only one curve is shown for the stiff condition, as it is kept constant.

scaling factor, was also employed in the work of [25]. The quantity $Z_r = \hat{p}_r/\hat{u}_r$ denotes the acoustic impedance at the reference plane, which is experimentally measured. Fig. 5 presents a comparison of G_ϕ under stiff and non-stiff conditions at a global equivalence ratio of $\bar{\phi} = 0.66$, with $g = 0.65$, a value that will be discussed in more detail later. In the stiff case, the model features a gain of unity and a constant phase of π, as only the air flow perturbations contribute to the generation of equivalence ratio fluctuations. In contrast, the non-stiff case shows a frequency-dependent response, featuring an excess gain and a peak around 200 Hz. This excess gain arises from pressure-driven fuel flow modulation. As the mean velocity increases, the resistance of the fuel injection hole also increases, as indicated by the factor $1/(JM_2C_d)$ discussed above, which leads to a decrease in the peak gain. The phase response also deviates from the stiff case, exhibiting a transition attributed to the phase difference between pressure and velocity fluctuations at the injector.

3.2. Transport of equivalence ratio fluctuations

We now investigate the equivalence ratio fluctuations at the burner outlet. This firstly aims to quantify the equivalence ratio fluctuations that reach the flame base, and secondly seeks to validate the equivalence ratio fluctuation generation model developed in the preceding section. To this end, the transfer function, T_ϕ, as defined in Eq. (1), is considered.

The turbulent diffusion and dispersion will attenuate the equivalence ratio fluctuations as they propagate from the fuel injector to the flame. To model this transport process, a one-dimensional convective–diffusive equation is used, following the formulation proposed in [26]:

$$\frac{\partial \phi'}{\partial t} + \bar{u}\frac{\partial \phi'}{\partial x} - \Gamma_{\text{eff}}\frac{\partial^2 \phi'}{\partial x^2} = 0, \tag{8}$$

where Γ_{eff} is an effective diffusivity that models the effect of turbulent diffusion and dispersion. From this equation, a mixing transfer function H_{mix} can be derived [51], which maps the equivalence ratio fluctuations from the injection location to the flame base:

$$H_{\text{mix}} = \frac{\hat{\phi}/\bar{\phi}}{\hat{\phi}_3/\bar{\phi}_3} = \exp\left[\frac{\text{Pe}}{2}\left(1 - \sqrt{1 + i2\pi \text{St}_T \frac{4}{\text{Pe}}}\right)\right], \tag{9}$$

where $\text{St}_T = fL_{\text{inj}}/\bar{u}$ is the transport Strouhal number, and $\text{Pe} = \bar{u}L_{\text{inj}}/\Gamma_{\text{eff}}$ is the Peclet number, which quantifies the relative intensities of convective to diffusive processes in the transport of equivalence ratio fluctuations. The transfer function T_ϕ becomes:

$$\begin{aligned} T_\phi &= G_\phi H_{\text{mix}} \\ &= g\left(\frac{-Z_r\bar{u}_r}{\bar{\rho}_n C_d \bar{u}_n^2} - 1\right)\exp\left[\frac{\text{Pe}}{2}\left(1 - \sqrt{1 + i2\pi \text{St}_T \frac{4}{\text{Pe}}}\right)\right]. \end{aligned} \tag{10}$$

In this formulation, g and Pe are the only unknown parameters that could be determined by fitting to the experimental measurements.

3.3. Comparison between models and experimental measurements

Fig. 6 compares the experimentally determined T_ϕ with model predictions under stiff and non-stiff conditions. The corresponding free parameters used in the model are summarized in Table 2. For the non-stiff condition, the parameters were determined via a least-squares fit to the experimental data. To facilitate comparison, the same Pe was applied to the stiff condition, while the modified scaling factor was fixed at $g = 1$.

Experimental results at a global equivalence ratio $\bar{\phi} = 0.66$ and varying airflow velocities $\bar{u}_2$ are shown in Fig. 6(a), and the results for a fixed airflow velocity $\bar{u}_2 = 11.6$ m/s over different global equivalence ratios $\bar{\phi}$ are presented in Fig. 6(b). In all cases, the gain exhibits a low-pass filter behavior, with a mild local maximum near 200 Hz, and nearly decays to zero beyond approximately 300 Hz, indicating that the mixing process attenuates high-frequency equivalence ratio fluctuations. Notably, the frequency region of the mild local maximum remains nearly unchanged across different $\bar{u}_2$ and $\bar{\phi}$. The phase of T_ϕ is characterized by a constant slope, indicating a convective time delay associated with the transport of equivalence ratio fluctuations from the injection point to the burner outlet. At frequencies beyond approximately 300 Hz, the phase measurements become less reliable as the gain approaches zero.

The model under the stiff condition shows good agreement with the experimental data in terms of both the phase and the overall trend of the gain, but it significantly underestimates the gain near 200 Hz. When the model is evaluated under the non-stiff condition, the mild local maximum around this frequency is successfully captured. This improvement is attributed to the inclusion of pressure-induced equivalence ratio fluctuations in the model formulation.

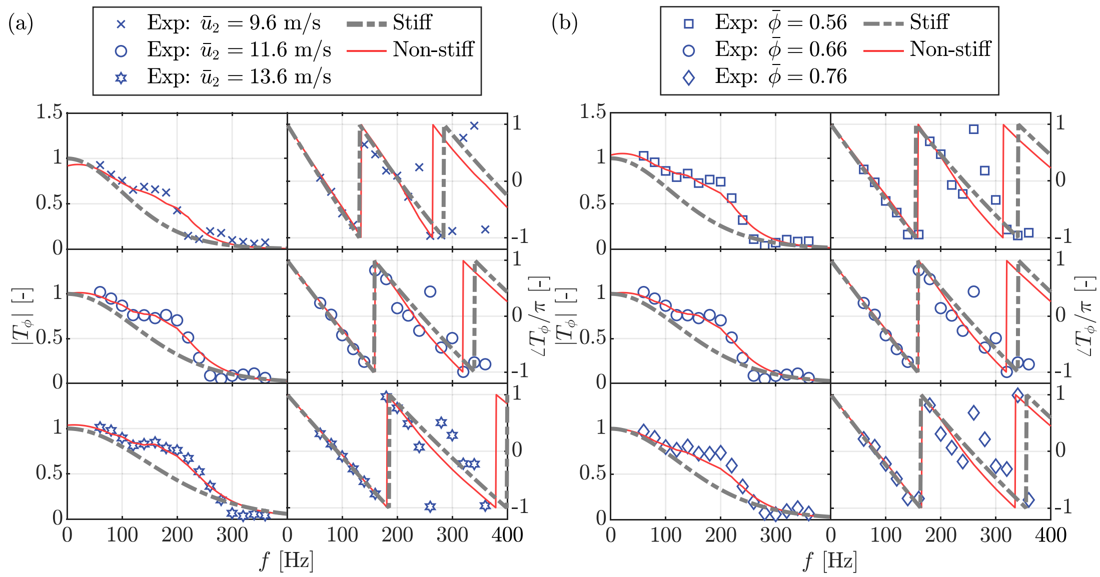

Fig. 6. Transfer functions of equivalence ratio fluctuations, T_ϕ, comparing experimental measurements with model predictions. (a) Results for varying air mass flow rates at a fixed global equivalence ratio, $\bar{\phi} = 0.66$. The corresponding free parameters are taken from cases C4, C2 and C5 (from top to bottom), as listed in Table 2. (b) Results for varying global equivalence ratios at a constant air mass flow rate, $\bar{V}_{\mathrm{Air}} = 21.70$ L/min. The corresponding free parameters are taken from cases C1, C2 and C3 (from top to bottom), as listed in Table 2.

Table 2
Selection of free parameters and key operating conditions for the non-stiff condition.

Parameters	C1	C2	C3	C4	C5
$\bar{\phi}$ [−]	0.56	0.66	0.76	0.66	0.66
$\bar{u}_2$ [m/s]	11.6	11.6	11.6	9.6	13.6
g [−]	0.62	0.66	0.70	0.62	0.66
Pe [−]	45.1	52.9	50.6	50.3	53.3

At the low-frequency limit, the model under both conditions predicts that the phase approaches π, which is consistent with the experimental observations. For the gain, the stiff condition leads to a value approaching unity, whereas under the non-stiff condition the gain exhibits a slight deviation from unity. This deviation depends on the acoustic impedance of the burner at the low-frequency limit.

It should be note that, in the model formulation, the acoustic impedance of the injector orifice is formulated under a low-frequency assumption, in which the acoustic reactance is neglected. While this approximation is appropriate in the low-frequency range of interest, it may contribute to additional phase deviations as the frequency increases. This limitation may contribute to the phase discrepancies observed in Fig. 6, where deviations between the modeled and measured phase become noticeable above approximately 240 Hz.

4. Flame transfer functions and flame dynamics

4.1. Flame shape and flame transfer functions

The next step is to examine how these equivalence ratio fluctuations influence the flame dynamics. We first examine the flame shape without modulation. Fig. 7(a) displays the time-averaged OH* chemiluminescence images and their corresponding Abel inversion for a global equivalence ratio $\bar{\phi} = 0.66$, across three different airflow velocities $\bar{u}_2$. Each image is normalized by its own maximum emission intensity. All flames exhibit a conical shape, with flame length increasing as the bulk velocity rises. Fig. 7(b) shows the time-averaged streamwise intensity profiles $\bar{I}_x$, obtained by integrating the emission intensity along the radial direction. These profiles are normalized by the maximum value across all three conditions. The results indicate that the maximum

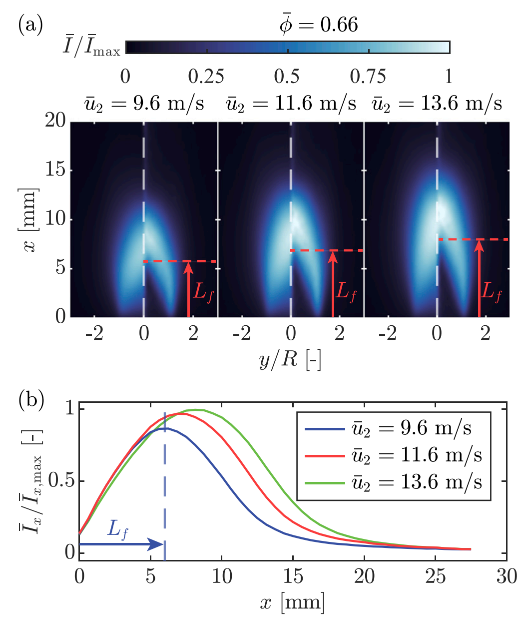

Fig. 7. (a) Time-averaged OH* emission (left), following an Abel inversion (right). Each image is normalized by its own maximum emission intensity. (b) Time-averaged streamwise intensity profiles $\bar{I}_x$, normalized by the maximum value across all three conditions. Both figures were operating at the same equivalence ratio ($\bar{\phi} = 0.66$) with different mean air flow velocities $\bar{u}_2$.

intensity increases as the bulk velocity rises. In this study, the axial location where $\bar{I}_x$ is maximum is chosen to denote the flame length L_f [32].

The flame dynamics are now investigated by analyzing the FTF, as defined in Eq. (2). Fig. 8 shows FTFs for nine different operating conditions. All FTF gains exhibit a global low-pass filter behavior. Superimposed on this trend is a clear gain modulation phenomenon, characterized by the presence of local maxima and minima. This modulation usually results from the constructive and destructive interference between two perturbations with different time delays. Notably, the gain

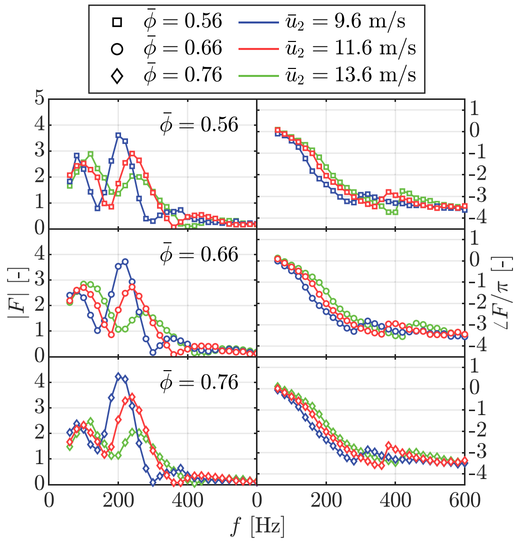

Fig. 8. Flame transfer functions F as a function of frequency.

curves are sensitive to the mean flow velocity. This is evidenced by the shifting positions of peaks and troughs as velocity varies. Furthermore, the magnitude of the second peak differs substantially across different mean flow velocities. A distinct phase jump is observed at the frequency corresponding to each local minimum in the gain, characterized by an abrupt change in the slope of the phase curve.

4.2. Rescaling and collapses of the FTFs

A central objective in the study of flame dynamics is to identify characteristic dimensionless parameters that capture the underlying physical mechanisms. In this context, scaling similar to those used by Moon et al. [21] are used. Fig. 9(a) shows that when the frequency in FTFs of Fig. 8 is non-dimensionalized by a convective time delay of equivalence ratio fluctuations, $\tau_c = (L_{\text{inj}} + L_f)/\bar{u}$, the resulting Strouhal number under reactive conditions, $\text{St}_r = f\tau_c$, effectively collapses both the phase trends and the position of peaks and troughs in the gain curves. The FTFs present a low-frequency limit of gain close to zero, and the fitted curve of phase shows a low-frequency limit close to $\pi/2$. These limits will be explained in more detail by using the MISO concept in the following discussions. This scaling confirms that this gain modulation arises predominantly from the constructive and destructive interference between velocity-induced and equivalence ratio-induced oscillations. However, the pronounced differences in the magnitude of the second peak is not captured by this scaling, suggesting that an additional mechanism influences its magnitude.

This behavior may be attributed to pressure-induced equivalence ratio fluctuations, whose presence has been confirmed through both experimental measurements and model predictions in Section 3. To better understand the influence of pressure-induced fluctuations, the overall OH* intensity fluctuation can be expressed as a linear superposition of two individual transfer functions, F_u and F_ϕ, corresponding to the flame response to velocity and equivalence ratio fluctuations at the burner outlet, respectively [52]. This results in a MISO FTF of the form:

$$\frac{\hat{I}}{\bar{I}} = F_u \frac{\hat{u}}{\bar{u}} + F_\phi \frac{\hat{\phi}}{\bar{\phi}}. \tag{11}$$

To relate the overall flame transfer function F to these individual transfer functions, the hydrodynamic effects upstream of the flame must be incorporated. For the present rig, large-scale vortical structures originating from the upstream swirler can be neglected due to the presence of a flow straightener upstream of the burner outlet, whereas equivalence ratio fluctuations must still be considered. Therefore, in the present systems with acoustically compact burners and non-stiff fuel injection, this yields the following expression:

$$F = F_u + F_\phi T_\phi. \tag{12}$$

As discussed in Section 3, the low-frequency behavior of the model under both conditions is close, indicating that the non-stiff effect of the present burner diminishes as the frequency approaches zero. In such cases, the low-frequency limit behavior should be similar to that of a partially premixed flame with a stiff injector, which has a low-frequency limit of zero gain [53]. If T_ϕ has a unity gain at zero frequency (which is approximately true in the present case, as shown in Fig. 6), the opposing effects of F_u and $F_\phi T_\phi$ do lead to complete cancellation, resulting in a zero gain and a phase of $\pi/2$. A qualitative analysis is conducted regarding the low-frequency limit of F, as shown in the supplementary material.

Alternatively, the system can be also formulated using the velocity fluctuations in the air and fuel supply lines as the two input variables. Under the non-stiff condition, these two inputs are intrinsically coupled, thereby reducing the system to a SISO representation. Accordingly, a SISO FTF can be constructed by selecting either the air-line or fuel-line velocity fluctuation as the reference input. Therefore, in the following analysis, an alternative FTF is defined by taking the fuel injection velocity u_n as the reference velocity:

$$F_{u_n} = \frac{\hat{I}/\bar{I}}{\hat{u}_n/\bar{u}_n}. \tag{13}$$

It should be noted that $\hat{u}_n$ is induced by incident pressure fluctuations and estimated using the acoustic impedance model of a circular aperture, as described in the previous section.

Fig. 9(b) presents the scaled results of F_{u_n} for different global equivalence ratios in different rows. As expected, the locations of the gain peaks and troughs collapse very well, reflecting the interference pattern that is dominated by the convective time delay τ_c as mentioned before. However, it is now more interesting to see that the gains of the FTFs at different air flow velocities $\bar{u}_2$ now collapse very well at each fixed $\bar{\phi}$ over the entire considered frequency range, especially near the second peak where large differences were seen in Fig. 9(a). This good collapse indicates that the pressure-induced equivalence ratio fluctuation is the main factor leading to the difference for the second peak in Fig. 9(a) — since the additional flame response driven by the pressure-induced equivalence ratio fluctuations is implicitly induced and normalized in the formulation of F_{u_n}, and the air-side velocity perturbation level is fixed in the experiment, the magnitude of gain now exhibits good collapse across different mean velocities. This collapse also provides indirect validation of the acoustic impedance model used to evaluate $\hat{u}_n$, which captures the pressure-driven contribution to the equivalence ratio fluctuations and thus the non-stiff effect. Regarding the phase response, the curves exhibit a consistent trend similar to that of $\angle F$, with a low-frequency limit close to zero, as $\hat{u}_r$ and $\hat{u}_n$ are out of phase at near-zero frequencies.

4.3. The spatiotemporal dynamics of OH* distribution

The underlying mechanism responsible for the gain modulation observed in these optical FTFs is further investigated using OH* chemiluminescence images. Although the spatial distribution of chemiluminescence intensity does not fully coincide with that of the heat release rate [54] — particularly in flames with mixture inhomogeneities [42] — it remains a qualitative indicator to provide spatially resolved insights into how perturbations in equivalence ratio and velocity affect different regions of the flame. First, the streamwise intensity profile I_x is used to analyze the spatiotemporal distribution along the axial direction. We consider a representative operating condition with $\bar{u}_2 = 9.6$ m/s, across different $\bar{\phi}$ and a range of forcing frequencies $f = 80, 140, 200, 300, 360$ Hz, corresponding to the local maxima and minima observed in the FTF gain, as shown in Fig. 8.

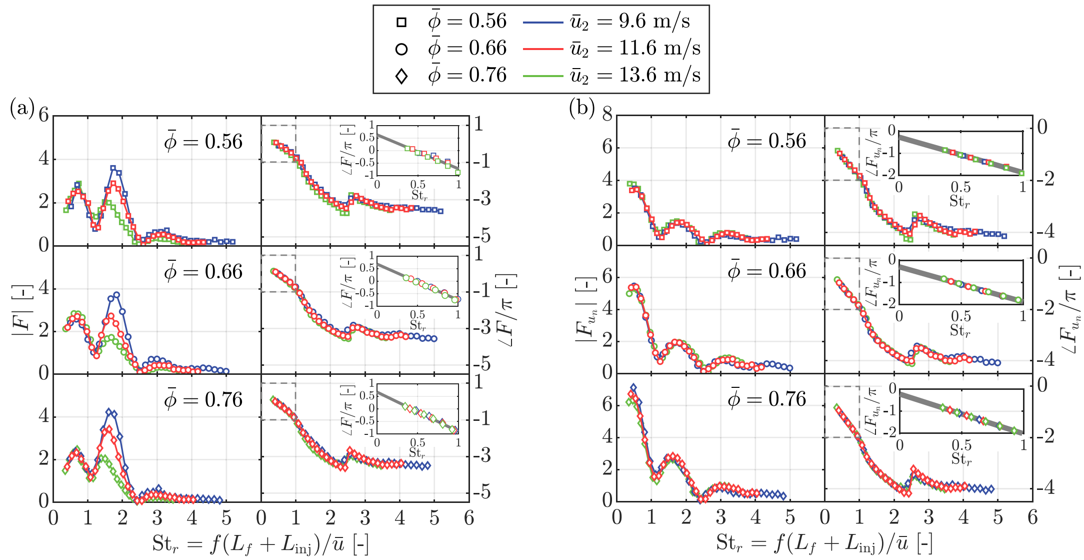

Fig. 9. Flame transfer functions: (a) F and (b) F_{u_n} as a function of the Strouhal number based on the convective time delay of equivalence ratio fluctuations. The fuel injection velocity u_n is estimated using the acoustic impedance model for a circular aperture. The gray solid line represents the fitted curve applied in the low-frequency range.

Fig. 10 shows the result of the phase-averaged I'_x over one modulation cycle. At $f = 80, 200$ and 360 Hz, only a single color is present along a constant t/T, indicating that the fluctuations along the entire flame front is nearly in phase. Such a distribution contributes positively or negatively to the overall OH* intensity fluctuations in a synchronized way, resulting in a local maximum in the FTF gain. At $f = 140$ and 300 Hz, a distinct color transition is observed along a constant t/T, implying the presence of streamwise destructive interference. This spatial cancellation results in a reduced overall intensity fluctuation, consistent with the local minimum in the FTF gain. Notably, the pattern orientation is nearly horizontal at $f = 80$, 140, and 200 Hz, indicating a predominantly non-convective mode behavior, where fluctuations do not seem to be convected along the flame. In contrast, at $f =$ 300 and 360 Hz, a slight inclination in the pattern emerges, suggesting a convective mode becomes gradually visible [55]. Nevertheless, spatial cancellation is still observed in the overall spatial integration of the OH* emission. This is attributed to the phase offset between the velocity-induced and equivalence ratio-induced oscillations, arising from their difference in convective time delay, given by $\Delta\tau = L_{\mathrm{inj}}/\bar{u}$. This phase difference is also evident in the results of the transfer function T_ϕ, which exhibits a phase close to zero at $f = 80$ and 200 Hz, indicating in-phase fluctuations, while approaching π at $f = 140$ and 300 Hz, suggesting out-of-phase behavior between the velocity and equivalence ratio fluctuations.

These observations suggest that the spatiotemporal distribution of OH* intensity can be broadly separated into two distinct regions — an upstream and a downstream region along the flame. To further investigate the different behaviors between these two regions and to verify whether the conclusion is consistent across various operating conditions, the flame emission images are divided into two interrogation windows: an upstream window (subscript "uw") and a downstream window (subscript "dw"). The cutting line is chosen at $x = L_f$, as illustrated in Fig. 10. A similar approach was adopted by Palies et al. [55], who divided the flame image into distinct spatial regions to analyze the interference behavior of a fully premixed swirling flame.

Fig. 11 presents the evolution of normalized OH* intensity for different windows across different forcing frequencies. At $f = 80$ and 200 Hz, the fluctuations from both windows is nearly in phase, leading to constructive interference and a stronger overall intensity fluctuation. In contrast, at $f = 300$ Hz, a clear phase offset between the upstream and downstream window signals leads to partial cancellation and a significant reduction in the overall fluctuation amplitude. These results are consistent with the streamwise interference patterns observed in Fig. 10. Notably, the amplitude of I'_{uw} decreases monotonically with frequency, indicating a progressively attenuated response in the upstream region, being consistent with the frequency-dependent damping seen in the equivalence ratio fluctuations. At higher frequencies ($f =$ 360 and 500 Hz), the overall intensity fluctuation is dominated by the downstream window, as evidenced by the close match between I' and I'_{uw}. Interestingly, the amplitude of I'_{dw} does not exhibit a monotonic trend with frequency, suggesting that the upstream and downstream regions of the flame possess distinct frequency-dependent response characteristics.

4.4. Discussions about the physical mechanism

To understand the different flame responses in the two windows, the same methodology used to characterize the overall flame response is now applied separately to each interrogation window. Two FTFs are introduced to quantify and compare the dynamic behavior of the upstream and downstream region of the flame, defined as:

$$F_{\mathrm{uw}}(\omega) = \frac{\hat{I}_{\mathrm{uw}}(\omega)/\bar{I}}{\hat{u}_r/\bar{u}_r}, \quad F_{\mathrm{dw}}(\omega) = \frac{\hat{I}_{\mathrm{dw}}(\omega)/\bar{I}}{\hat{u}_r/\bar{u}_r}. \tag{14}$$

Fig. 12(a) shows the FTF of the downstream window, F_{dw}. A clear gain modulation is observed across all operating conditions, with peaks and troughs that follow the trends seen in the overall FTFs. The phase response displays a globally monotonic decrease with frequency, along with distinct phase jumps near local gain minima. When frequency approaches zero, the gain of F_{dw} approaches zero and the fitted curve of phase tends toward $\pi/2$. These results suggest that the flame response in the downstream region is strongly influenced by the temporal interference between velocity-induced and equivalence ratio-induced oscillations.

Fig. 12(b) presents the FTF of the upstream window, F_{uw}. In contrast to F and F_{dw}, the behavior of F_{uw} deviates significantly. Rather than exhibiting a modulation pattern with multiple distinct peaks and troughs, F_{uw} displays a classical low-pass filter characteristic. The phase response decreases linearly with frequency, with a slope corresponding to the convective time delay associated with the transport of equivalence ratio fluctuations. The low-frequency limit of the phase is close to π, consistent with the quasi-steady behavior of equivalence ratio fluctuations. The rapid attenuation of gain in the high-frequency range is also

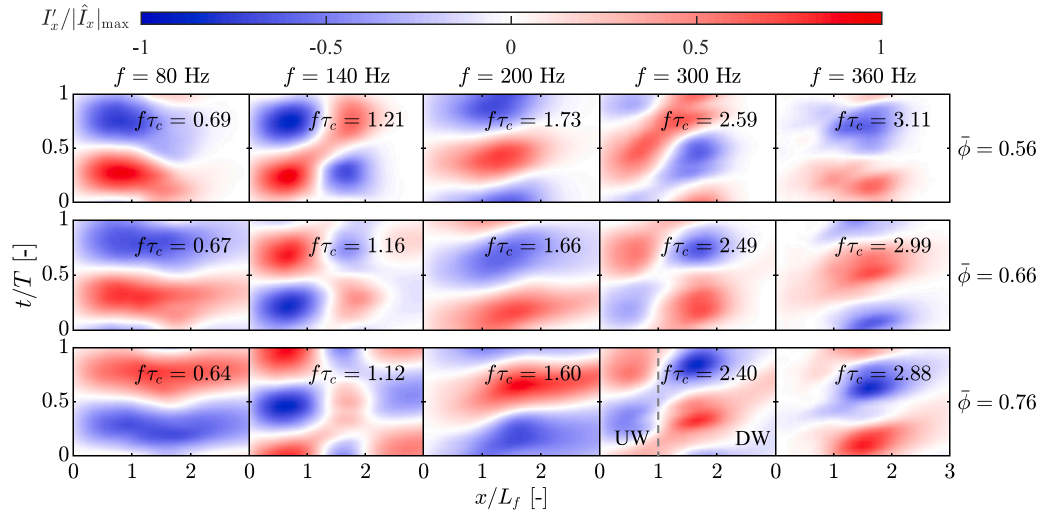

Fig. 10. Evolution of phase-averaged streamwise OH* intensity fluctuation I'_x. Each column is normalized by the maximum value of respective column. Operating condition: $\bar{\phi} = 0.56, 0.66, 0.76$, $\bar{u}_2 = 9.6$ m/s. Each column illustrates the spatiotemporal distribution of the fluctuations for selected forcing frequencies corresponding to local maxima and minima in the FTF gain. Gray dash line is an example of cutting line.

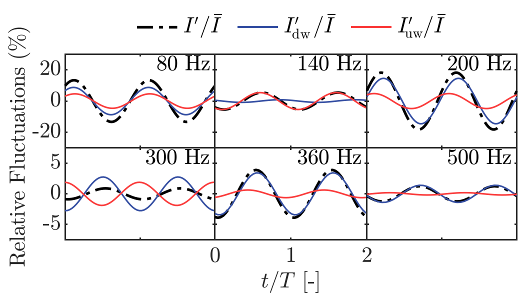

Fig. 11. Time evolution of OH* intensity extracted from the upstream (I'_{uw}) and downstream (I'_{dw}) interrogation windows, compared with the overall intensity signal (I'). All signals are extracted via Fourier transform at the forcing frequency and normalized by the time-averaged values ($\bar{I}$). Operating condition: $\bar{\phi} = 0.66$, $\bar{u}_2 = 9.6$ m/s.

in line with earlier observations from the phase-averaged streamwise intensity profiles. These observations indicate that the upstream region is predominantly influenced by equivalence ratio fluctuations.

These results allow for a deeper physical interpretation of the contrasting behaviors observed in the upstream and downstream windows, which can be associated with the dynamics of the flame base and flame tip, respectively. In a perturbed flow — whether due to velocity or equivalence ratio modulation — flame wrinkles are generated at the burner exit and convect downstream along the flame front. These wrinkles modulate the flame surface area, leading to substantial variations in heat release rate and thus in OH* intensity. Additionally, OH* intensity is directly influenced by the local equivalence ratio [45, 46]. Therefore, monitoring the flame surface area variation, rather than OH* intensity fluctuations alone, offers useful additional information for evaluating the superposed effects of velocity- and equivalence ratio-induced oscillations.

Fig. 13 presents the time evolution of the number of flame-occupied pixels in different regions. The fluctuations in the upstream window (NP'_{uw}) remain consistently small, whereas a close correspondence between NP'_{dw} and NP' is observed. This suggests that the overall surface area variations are dominated by the downstream window. As a result, the intensity fluctuations in the downstream window reflect the combined influence of velocity- and equivalence ratio-induced oscillations, manifesting as the modulation pattern observed in F_{dw}. Conversely, the upstream window exhibits only small surface deformation. It should be noted that, within the studied frequency range, the convective wavelength $\bar{u}/f$ is considerably larger than the flame length, yielding a bulk motion that stretches the conical-shaped flame without inducing localized wrinkles. Consequently, surface area changes in the upstream window are negligible, and OH* intensity variations in this region are primarily driven by fluctuations in local equivalence ratio. Accordingly, the behavior of F_{uw} is dominated by equivalence ratio perturbation dynamics.

Although the present spatiotemporal analysis does not isolate the non-stiff injection effect as directly as the previous analysis, it remains essential for understanding its physical implications. The non-stiff injection primarily influences the generation of equivalence ratio fluctuations, which in turn govern the flame response. Therefore, analyzing the spatial distribution of OH* chemiluminescence under equivalence ratio perturbations provides important insight into how these fluctuations affect the flame dynamics. In this context, F_{dw} exhibits a response that closely resembles the overall FTF, including the additional gain modulation associated with the non-stiff effect. This indicates that the influence of non-stiff injection is more pronounced in the downstream region, where the combined effect of velocity and equivalence ratio fluctuations determines the flame response.

5. Conclusion

The present study explored the dynamic response of a partially premixed pure hydrogen flame in a micro-mix burner with non-stiff fuel injection. The following key conclusions are drawn.

In a non-stiff fuel injection system, equivalence ratio fluctuations arise from both air-stream velocity perturbations and pressure-driven modulation of the fuel flow rate. The result of a predictive model shows that neglecting the pressure-driven contribution leads to an underestimation of the fluctuation amplitude, whereas the non-stiff formulation captures the measured gain. This demonstrates that pressure-induced equivalence ratio fluctuations are a key mechanism in systems with non-stiff injection.

The flame response was determined by FTF based on OH* chemiluminescence intensity. The FTFs exhibit a gain modulation pattern resulting from constructive and destructive interference between velocity- and equivalence ratio-induced oscillations. A notable amplification in the gain, particularly near the second peak, is observed and cannot be fully collapsed using a Strouhal number scaling. This deviation is found to be due to pressure-induced equivalence ratio fluctuations. To isolate this effect, an alternative FTF is introduced, taking the fuel injection

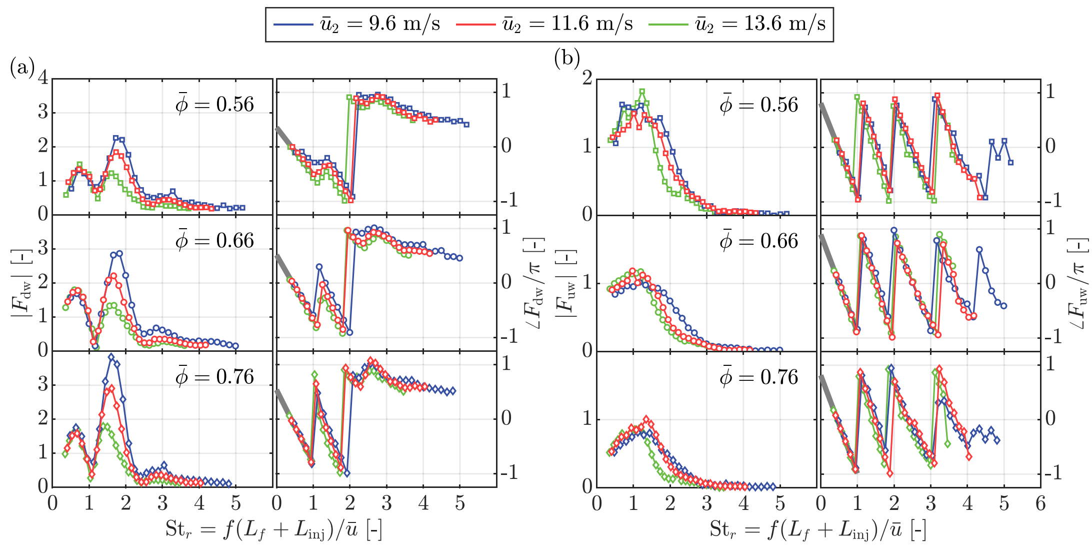

Fig. 12. Flame transfer functions F_{dw} and F_{uw} plotted as a function of the Strouhal number St_r. The gray solid line represents a fitted curve applied in the low-frequency range.

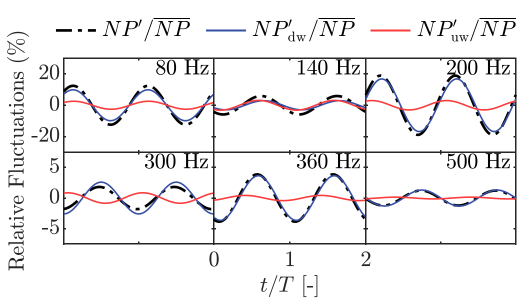

Fig. 13. Time evolution of the number of flame-occupied pixels in the upstream (NP'_{uw}) and downstream interrogation windows (NP'_{dw}), compared with the overall intensity signal (NP'). All signals are extracted via Fourier transform at the forcing frequency and normalized by the time-averaged values ($\overline{NP}$). Operating condition: $\bar{\phi} = 0.66$, $\bar{u}_2 = 9.6$ m/s.

velocity as the reference input. This approach effectively collapses the FTF gains and confirms the influence of pressure-driven equivalence ratio fluctuations.

The spatiotemporal distribution of OH* intensity revealed distinct flame dynamic behaviors of the upstream and downstream region when subjected to modulations in velocity and equivalence ratio perturbations. The upstream region responds predominantly to equivalence ratio fluctuations, exhibiting a low-pass filter behavior. In contrast, the downstream region is influenced by both velocity- and equivalence ratio-induced effects, resulting in pronounced interference patterns. The phase difference between these regions produces the observed gain modulation in the optical-based FTFs and shapes the overall unsteady flame dynamics.

It should be noted that the present study is based on a single-nozzle configuration. Collective effects present in practical multi-nozzle micromix combustors, such as flame–flame interaction and transverse acoustic modes, are therefore not captured. The results should thus be interpreted as nozzle-scale insight, providing a foundation for future investigations and modeling efforts addressing more complex multi-nozzle and system-level configurations. Nevertheless, under conditions where the burner can be considered acoustically compact, the modeling framework and methodology developed in this work can be extended to more complex burner configurations.

CRediT authorship contribution statement

Weiyuan Liang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhixin Zhao:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Chengxi Miao:** Writing – review & editing. **Bowen Sa:** Writing – review & editing. **Bo Zhou:** Writing – review & editing, Resources, Methodology, Conceptualization. **Dong Yang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.energy.2026.140827.

Data availability

Data will be made available on request.

References

[1] Lefebvre AH, Ballal DR. Gas turbine combustion: alternative fuels and emissions. CRC Press; 2010.

[2] Masri AR. Challenges for turbulent combustion. Proc Combust Inst 2021;38(1):121–55.

[3] Noble D, Wu D, Emerson B, Sheppard S, Lieuwen T, Angello L. Assessment of current capabilities and near-term availability of hydrogen-fired gas turbines considering a low-carbon future. J Eng Gas Turbines Power 2021;143(4):041002.

[4] Griffiths S, Sovacool BK, Kim J, Bazilian M, Uratani JM. Industrial decarbonization via hydrogen: A critical and systematic review of developments, socio-technical systems and policy options. Energy Res Soc Sci 2021;80:102208.
[5] Capurso T, Stefanizzi M, Torresi M, Camporeale S. Perspective of the role of hydrogen in the 21st century energy transition. Energy Convers Manage 2022;251:114898.
[6] Kalantari A, McDonell V. Boundary layer flashback of non-swirling premixed flames: Mechanisms, fundamental research, and recent advances. Prog Energy Combust Sci 2017;61:249–92.
[7] Ranjan R, Clemens NT. Insights into flashback-to-flameholding transition of hydrogen-rich stratified swirl flames. Proc Combust Inst 2021;38(4):6289–97.
[8] Aniello A, Poinsot T, Selle L, Schuller T. Hydrogen substitution of natural-gas in premixed burners and implications for blow-off and flashback limits. Int J Hydrog Energy 2022;47(77):33067–81.
[9] Cuoci A, Frassoldati A, Cozzi F. Numerical predictions of flashback limits of H2-enriched methane/air premixed laminar flames. Proc Combust Inst 2024;40(1–4):105696.
[10] Flebbe S, Tartsch S, Sattelmayer T. Thermal runaway effect of the centerbody on boundary layer flashback in a swirl-stabilized gas turbine burner operated with hydrogen. Int J Hydrog Energy 2024;61:6–12.
[11] Funke H-W, Beckmann N, Abanteriba S. An overview on dry low NOx micromix combustor development for hydrogen-rich gas turbine applications. Int J Hydrog Energy 2019;44(13):6978–90.
[12] York WD, Ziminsky WS, Yilmaz E. Development and testing of a low NOx hydrogen combustion system for heavy-duty gas turbines. J Eng Gas Turbines Power 2013;135(2):022001.
[13] Asai T, Dodo S, Karishuku M, Yagi N, Akiyama Y, Hayashi A. Performance of multiple-injection dry low-NOx combustors on hydrogen-rich syngas fuel in an IGCC pilot plant. J Eng Gas Turbines Power 2015;137(9):091504.
[14] Chen X, Wang H, Wang X, Liu X, Zhu Y. Fuel/air mixing characteristics of a Micromix burner for hydrogen-rich gas turbine. Energy 2023;282:128786.
[15] Lopez-Ruiz G, Alava I, Blanco J. Impact of H2/CH4 blends on the flexibility of micromix burners applied to industrial combustion systems. Energy 2023;270:126882.
[16] Chen M, Zhang L, Xing C, Bao Y, Qiu P, Zhang W, Sun S, Zhao Y. Experimental and numerical simulation study of the effect of mixing on the characteristics of swirl/non-swirl micromix flames. Energy 2024;307:132570.
[17] Qi H, Tian X, Feng Z, Yang Y, Liu D, Wang Q, Wang G, Xu L, Xia X. Investigation of longitudinal self-excited combustion instability in a micromix hydrogen combustor. Proc Combust Inst 2024;40(1–4):105762.
[18] Ben Y, Xiong Y, Liu Z, Yang N, Zhao L, Xu X. Measurement of flame describing function of jet multinozzle array combustor and prediction of its thermoacoustic oscillation frequency. J Eng Gas Turbines Power 2025;147(11).
[19] Kroniger D, Horikawa A, Funke HH-W, Pfaeffle F, Kishimoto T, Okada K. Experimental and numerical investigation on the effect of pressure on micromix hydrogen combustion. In: ASME conference proceedings. GT2021-58926.
[20] Kroniger D, Horikawa A, Okada K, Ashida Y. Novel fuel injector geometry for enhancing the fuel flexibility of a dry low NOx micromix flame. In: ASME conference proceedings. GT2022-83025.
[21] Moon K, Martin R, Schuermans B, Noiray N. Transfer functions of lean fully-and technically-premixed jet-stabilized turbulent hydrogen flames. Proc Combust Inst 2024;40(1–4):105256.
[22] Morgans AS, Yang D. Thermoacoustic instability in combustors. Annu Rev Fluid Mech 2025;57:9–33.
[23] Huang Y, Yang V. Dynamics and stability of lean-premixed swirl-stabilized combustion. Prog Energy Combust Sci 2009;35(4):293–364.
[24] Lieuwen T, Zinn BT. The role of equivalence ratio oscillations in driving combustion instabilities in low NOx gas turbines. Symp (Int) Combust 1998;27(2):1809–16.
[25] Peracchio AA, Proscia WM. Nonlinear heat-release/acoustic model for thermoacoustic instability in lean premixed combustors. J Eng Gas Turbines Power 1999;121(3):415–21.
[26] Bluemner R, Paschereit CO, Oberleithner K. Generation and transport of equivalence ratio fluctuations in an acoustically forced swirl burner. Combust Flame 2019;209:99–116.
[27] Zhang G, Yang N, Wang Y, Li L, Wang X, Yang D, Lin Y, Sun X. Combustion instabilities of bluff-body stabilized liquid spray flames coupled with pressure oscillations in the fuel line. Combust Flame 2025;273:113911.
[28] Huber A, Polifke W. Impact of fuel supply impedance on combustion stability of gas turbines. In: ASME conference proceedings. GT2008-51193.
[29] Kim KT, Lee JG, Quay BD, Santavicca D. Response of partially premixed flames to acoustic velocity and equivalence ratio perturbations. Combust Flame 2010;157(9):1731–44.
[30] Komarek T, Polifke W. Impact of swirl fluctuations on the flame response of a perfectly premixed swirl burner. J Eng Gas Turbines Power 2010;132(6):061503.
[31] Palies P, Durox D, Schuller T, Candel S. Nonlinear combustion instability analysis based on the flame describing function applied to turbulent premixed swirling flames. Combust Flame 2011;158(10):1980–91.
[32] Æsøy E, Aguilar JG, Wiseman S, Bothien MR, Worth NA, Dawson JR. Scaling and prediction of transfer functions in lean premixed H2/CH4-flames. Combust Flame 2020;215:269–82.
[33] Briones AM, Aggarwal SK, Katta VR. Effects of H2 enrichment on the propagation characteristics of CH4–air triple flames. Combust Flame 2008;153(3):367–83.
[34] Kuhlmann J, Marragou S, Boxx I, Schuller T, Polifke W. LES-based prediction of technically premixed flame dynamics and comparison with perfectly premixed mode. Phys Fluids 2022;34(8).
[35] Garcia AM, Le Bras S, Prager J, Boxx I, Polifke W. Impact of H2-enrichment on the response of a partially premixed CH4–air flame to velocity and equivalence ratio fluctuations. Combust Flame 2024;268:113595.
[36] Schuermans B, Bellucci V, Guethe F, Meili Fo, Flohr P, Paschereit CO. A detailed analysis of thermoacoustic interaction mechanisms in a turbulent premixed flame. In: ASME conference proceedings. GT2004-53831.
[37] Ćosić B, Terhaar S, Moeck JP, Paschereit CO. Response of a swirl-stabilized flame to simultaneous perturbations in equivalence ratio and velocity at high oscillation amplitudes. Combust Flame 2015;162(4):1046–62.
[38] Paschereit CO, Schuermans B, Polifke W, Mattson O. Measurement of transfer matrices and source terms of premixed flames. J Eng Gas Turbines Power 2002;124(2):239–47.
[39] Li C, Su J, Yang D. The acoustics of inclined openings with a low-Mach mean flow. J Sound Vib 2025;119324.
[40] Mellqvist J, Rosén A. DOAS for flue gas monitoring—I. Temperature effects in the UV/visible absorption spectra of NO, NO2, SO2 and NH3. J Quant Spectrosc Radiat Transfer 1996;56(2):187–208.
[41] Zhao Z, Wang Z, Sun M, Chen H, Yang D, Zhou B. Flame describing function of conical laminar premixed flames subjected to parasite-velocity decoupled equivalence ratio oscillation. Combust Flame 2025;275:114078.
[42] Lauer M, Sattelmayer T. On the adequacy of chemiluminescence as a measure for heat release in turbulent flames with mixture gradients. J Eng Gas Turbines Power 2010;132(6):061502.
[43] Kang H, Yoon C, Kim KT. Experimental and numerical investigations of forced response of multi-element lean-premixed hydrogen flames. Combust Flame 2023;258:113079.
[44] Kruljevic B, Darabiha N, Durox D, Vaysse N, Renaud A, Vicquelin R, Fiorina B. Experimentation and simulation of a swirled burner featuring cross-flow hydrogen injection with a focus on the OH* chemiluminescence. Combust Flame 2025;273:113945.
[45] Higgins B, McQuay M, Lacas F, Rolon J-C, Darabiha N, Candel S. Systematic measurements of OH chemiluminescence for fuel-lean, high-pressure, premixed, laminar flames. Fuel 2001;80(1):67–74.
[46] Schuller T, Marragou S, Oztarlik G, Poinsot T, Selle L. Influence of hydrogen content and injection scheme on the describing function of swirled flames. Combust Flame 2022;240:111974.
[47] Wang G, Faure-Beaulieu A, Schuermans B, Noiray N. Flame transfer function analysis of hydrogen diffusion swirl flames. Proc Combust Inst 2024;40(1–4):105727.
[48] Cummings A, Eversman W. High amplitude acoustic transmission through duct terminations: Theory. J Sound Vib 1983;91(4):503–18.
[49] Bellucci V, Flohr P, Paschereit CO, Magni F. On the use of Helmholtz resonators for damping acoustic pulsations in industrial gas turbines. J Eng Gas Turbines Power 2004;126(2):271–5.
[50] Lichtarowicz A, Duggins RK, Markland E. Discharge coefficients for incompressible non-cavitating flow through long orifices. J Mech Eng Sci 1965;7(2):210–9.
[51] Moeck J, Bothien M, Schimek S, Lacarelle A, Paschereit C. Subcritical thermoacoustic instabilities in a premixed combustor. In: 14th AIAA/CEAS aeroacoustics conference (29th AIAA aeroacoustics conference). 2008, p. 2946.
[52] Huber A, Polifke W. Dynamics of practical premixed flames, part I: model structure and identification. Int J Spray Combust Dyn 2009;1(2):199–228.
[53] Polifke W, Lawn C. On the low-frequency limit of flame transfer functions. Combust Flame 2007;151(3):437–51.
[54] Schiavone FG, Torresi M, Camporeale SM, Laera D. Impact of the operating conditions on the OH* distribution and its correlation with the heat release rate in hydrogen–air flames. Combust Flame 2025;282:114452.
[55] Palies P, Durox D, Schuller T, Candel S. The combined dynamics of swirler and turbulent premixed swirling flames. Combust Flame 2010;157(9):1698–717.